

Mitchell C. Bosley

CENIA - National Center for Artificial Intelligence
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Google Scholar

Research Profile

I am a political scientist and artificial intelligence researcher studying how machine learning and large language models reshape democratic institutions, public policy, and political behavior. My research program is driven by the conviction that artificial intelligence will transform how humans acquire and interpret political information and deserves to be studied as a political phenomenon in its own right. To this end, my work focuses on three areas: (1) developing AI-driven measurement techniques to analyze political institutions and political discourse; (2) understanding how AI-human interaction shapes political attitudes and behavior; and (3) using AI-powered simulations to gain insight into political processes. Across these projects, I combine causal inference, text-as-data, and large-scale experiments with AI systems, and my work on AI and politics has been published in venues such as the *American Political Science Review* and the *Journal of Political Institutions and Political Economy*, with ongoing work on AI-mediated prejudice reduction under revise-and-resubmit at *PNAS Nexus*.

Academic Appointments

Postdoctoral Researcher, CENIA - National Center for Artificial Intelligence
Santiago, Chile, 2025 - Present

Postdoctoral Researcher, Schwartz Reisman Institute for Technology and Society
University of Toronto, 2024 - 2025

Education

University of Michigan

Ph.D., Political Science, 2024 (awarded September 2024)

Fields: *Political Methodology, Comparative Politics, Political Economy*

Dissertation: *Three Papers in the Applied Use of Machine Learning and Artificial Intelligence Models for the Analysis of Political Text Data*

Committee: Walter Mebane (Co-Chair), George Tsebelis (Co-Chair), Yuki Shiraito, Ken Kollman, Charles Shipan

University of British Columbia

M.A., Political Science, 2018

B.A., Political Science (Honors Program), 2016

Publications and Works in Progress

Peer-Reviewed Publications

- *Improving Probabilistic Models in Text Classification via Active Learning*
American Political Science Review
with Saki Kuzushima, Yuki Shiraito, and Ted Enamorado
- *Towards Qualitative Measurement at Scale: A Prompt-Engineering Framework for Large-Scale Analysis of Deliberative Quality in Parliamentary Debates*
Journal of Political Institutions and Political Economy, Special Issue on AI and Political Institutions

Under Review and Revise & Resubmit

- *Can AI Help Reduce Prejudice? Evaluating the Effectiveness of AI-Powered Personalized Persuasion on Support for Transgender Rights*
Revise & Resubmit, **PNAS Nexus**
with John Holbein, Charles Crabtree, and Semra Sevi

Works in Progress

1. Text Analysis and Legislative Research

- *From Debate to Statute: Tracing Legislative Influence Using Generative AI*
with Julia Park and Kevin Esterling
- *Reform Velocity in Latin America: Measuring Policy Change in IMF Article IV Reports with Large Language Models*
with collaborators at the Inter-American Development Bank
- *Using LLMs to Evaluate Evidence Support for Grand Theories of Politics at Scale*
with Tore Wig

2. AI-Powered Persuasion

- *AI Chatbots and Public Support for Disability Welfare Policy*
with Jac Larner and Joshua Thorpe

3. Digital Twins and Political Behavior Simulation

- *Building Digital Twins from Interviews: Do AI Interviewers Match Human Fieldwork?*
with Juan Pablo Luna, Sergio Toro, Naim Bro, and Andrés Abeliuk

Skills

Quantitative Methods

Causal inference; experimental and survey design; machine learning; natural language processing; text-as-data

AI Engineering

Design, evaluation, and maintenance of LLM-based systems for chatbots, measurement pipelines, and policy simulations; prompt and evaluation design for agentic workflows

Programming and Computation

Python, R, JavaScript, high performance computing (Slurm), version control (Git)

Languages

English (Native), French (Intermediate), Spanish (Intermediate)

Teaching Experience

As Instructor of Record

Pontificia Universidad Católica de Chile

Applied Machine Learning for Social Science (scheduled March 2026)

Planned course in the Department of Sociology focused on applied use of machine learning tools for social research

University of British Columbia

Investigating Politics: An Introduction to Scientific Political Science (2023)

Full responsibility for course design, lectures, assignments, and grading

As Graduate Student Instructor

University of Michigan and University of British Columbia

Introduction to Game Theory; Research Ethics for Graduate Students; Introduction to World Politics; Introduction to Comparative Politics; Introduction to Canadian Politics

Awards

- **Rackham Doctoral Fellowship**, University of Michigan

- 2024: \$3,000 USD
- **Department of Political Science Director of Graduate Studies Award**, University of Michigan
 - 2023: \$1,500 USD
- **Gerald R. Ford Dissertation Fellowship**, University of Michigan
 - 2023-2024: \$25,980 USD
 - 2022-2023: \$24,054 USD
- **Text as Data Association Conference Attendance Grant**
 - University of Massachusetts, Amherst, 2023: \$530 USD
 - Cornell Tech, 2022: \$1,600 USD
- **Empirical Implications of Theoretical Models (EITM) Group Attendance Grant**, University of California, Berkeley
 - 2022: \$900 USD
- **MapleMeth (Canadian PolMeth) Attendance Grant**, University of Western Ontario
 - 2023: \$800 CAD
 - 2022: \$1,100 CAD
- **Weiser Center for Emerging Democracies Research Grant**, University of Michigan
 - 2021: \$4,000 USD
- **Summer Collaboration Grant**, University of Michigan
 - 2021: \$3,000 USD
- **Rackham Doctoral Fellowship**, University of Michigan
 - 2018-2019: \$24,000 USD
- **Rackham MPSA Travel Fund**, University of Michigan
 - 2023: \$500 USD
 - 2022: \$500 USD
- **Conference Travel Fund**, University of Michigan
 - PolMeth Conference, 2023: \$600 USD
 - American Political Science Association, 2023: \$600 USD
 - American Political Science Association, 2022: \$600 USD
 - Canadian Political Science Association, 2019: \$800 USD

Conference Presentations

2025

- *Institutional Change After Franchise Expansion: Evidence From British India, 1919-1947*, American Political Science Association.
- *From Debate to Statute: Tracing Legislative Influence Using Generative AI*, American Political Science Association.
- *Can AI Help Reduce Prejudice? Evaluating the Effectiveness of AI-Powered Personalized Persuasion on Support for Transgender Rights*, American Political Science Association.
- *Can AI Help Reduce Prejudice? Evaluating the Effectiveness of AI-Powered Personalized Persuasion on Support for Transgender Rights*, Society for Political Methodology (PolMeth).
- *Can AI Help Reduce Prejudice? Evaluating the Effectiveness of AI-Powered Personalized Persuasion on Support for Transgender Rights*, Midwestern Political Science Association.
- *Can AI Help Reduce Prejudice? Evaluating the Effectiveness of AI-Powered Personalized Persuasion on Support for Transgender Rights*, Conference on Social Media & Democratic Practice, Hoover Institution, Stanford University.
- *Building Digital Twins from Interviews: Do AI Interviewers Match Human Fieldwork?*, Latin American PolMeth (Poster).

2024

- *Towards Qualitative Measurement at Scale: A DQI Measurement Framework using Prompt-Engineering*, American Political Science Association.
- *Towards Qualitative Measurement at Scale: A DQI Measurement Framework using Prompt-Engineering*, Society for Political Methodology (PolMeth, UC Riverside, Poster).
- *Towards Qualitative Measurement at Scale: A DQI Measurement Framework using Prompt-Engineering*, JPIPE Conference on AI for the Study of Political Institutions (Paper).
- *Can AI Help Reduce Prejudice? Evaluating the Effectiveness of AI-Powered Personalized Persuasion on Support for Transgender Rights*, Toronto-Montreal Political Behavior Workshop (McGill University).
- *Can AI Help Reduce Prejudice? Evaluating the Effectiveness of AI-Powered Personalized Persuasion on Support for Transgender Rights*, Latin American PolMeth.

2023

- *When do the Rules Matter? Analyzing the Effect of Rule Change on Obstruction in Legislatures*, American Political Science Association.
- *Improving Probabilistic Models in Text Classification via Active Learning*, Society for Political Methodology (PolMeth).

- *Using Language Models to Measure Legislative Obstruction in the Canadian House of Commons*, Canadian Political Science Association.
- *Using Language Models to Measure Legislative Obstruction in the Canadian House of Commons*, Canadian Political Methodology Society (MapleMeth).
- *Do we still need BERT in the age of GPT? Comparing the benefits of domain-adaptation and in-context-learning approaches to using LLMs for Political Science Research*, Midwestern Political Science Association.
- *Measuring Obstructionary Speech in Legislatures using Large Language Models*, Text as Data Association Conference (Poster).

2022

- *When do the Rules Matter? Analyzing the Effect of Rule Change on Obstruction in Legislatures*, American Political Science Association (Poster).
- *Institutional Change After Franchise Expansion: Evidence From British India*, Midwestern Political Science Association.
- *When do the Rules Matter? Analyzing the Effect of Rule Change on Obstruction in Legislatures*, Midwestern Political Science Association.
- *Towards LegisBERT: Improving Language Models for the Analysis of Legislative Speech*, Canadian Political Methodology Society (MapleMeth).
- *Towards LegisBERT: Improving Language Models for the Analysis of Legislative Speech*, Text as Data Association Conference (Poster).

2021

- *Improving Probabilistic Models in Text Classification via Active Learning*, Midwestern Political Science Association.
- *Improving Probabilistic Models in Text Classification via Active Learning*, Text as Data Association Conference (Poster).

2020 and earlier

- 2020: *Improving Probabilistic Models in Text Classification via Active Learning*, American Political Science Association.
- 2019: *Rationalizing Legislative Rule Change: Three Cases of Rule Change in the Canadian House of Commons*, Canadian Political Science Association.

Research Methods

Quantitative Methods

Machine Learning, Natural Language Processing, Text Analysis, Causal Inference, Survey Experiments

Programming and Computation

R, Python, High Performance Computing (Slurm), Version Control (Git)

Languages

English (Native), French (Intermediate), Spanish (Intermediate)